On Existence of Digital Measure Constant

F. G. Ersezer, *Student Member, IEEE*

***Abstract*—Unidentified continuous functions are a generalized classification for all signals. The author assumes that any unidentified continuous function (any signal) is either digital (non-Archimedean continuum) or analog (Archimedean continuum), and gives a method to indicate a distinction between both continuum models. The difference between a value x and its digital version is the fractional part $\{x\}$, whose graph is Luzin's saw. Measuring this displacement relative to x with the scale-invariant measure dx/x, we show that the digital measure constant is $Hyp_1 = 1-\gamma$, where γ is the Euler–Mascheroni constant. The value does not depend on the step dx, and it is the same for every infinite upper bound, which is required because ordinary numbers are infinite at the scale dx. Over one unit length containing Hyp_0 steps, $Hyp_1 = (1-\gamma) - 1/(2Hyp_0)$ to leading order. Applied to the photon energy levels $E_n = E_1/n$, the digital spectrum falls short of the continuous one by $(1-\gamma)E_1$ in a continuum and by $(1-\gamma)E_1 - \frac{1}{2}E_{\min}$ in a discrete world, which gives a test between the two continuum models.**

***Index Terms*—Time-frequency analysis, non-standard analysis, digital topology**

I. Introduction

THE article "Approaches to analysis with infinitesimals following Robinson, Nelson and others" [18] has the section "The CD+II mindset". The terms of that section will be regarded, since they are foundations of set-theoretical identifications that are based on an "intuitively equal" algebraic partition (arithmetization) of the real number line (all real numbers) into a countably infinite amount of interval groups. (The work of Sophus Lie is a general reference for the historical viewpoint of continuous symmetric groups [2]–[4].)

"The Cantor–Dedekind postulate (CD) identifying the line in physical space with the real number line" (Cantor's main work is [5]; his arguments on reals and infinitesimals are on pages 26–55 of [17]).

"The Intended Interpretation hypothesis (II), entailing an identification of a standard $\mathbb{N}$ in its set-theoretic context, on the one hand, with ordinary intuitive counting numbers, on the other."

These main arguments will be the building blocks for the separation of unidentified continuous functions (generalized signals). The mathematical foundations are generally regarded whenever there is a need for deductive (Fourier or harmonic) functional analysis (see [20]). The author will regard non-standard analysis, since the main arguments for a digital continuum can be founded via inductive (instantaneous) non-standard notions of analysis. Rather than concerning applications about the statistical large-scale construction of signals on the real numbers to generate a generalized theory, the author establishes a deterministic model of small-scale construction for signals on $\mathbb{R}^*$ as a generalized theory and applies it to larger scales.

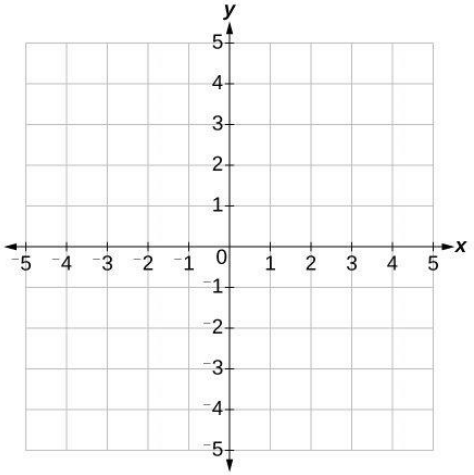


Fig. 1. Cartesian Coordinate System

The Cartesian Coordinate System is mainly a product of the algebraic partition of the real number line into "intuitively equal", countable and infinitely many interval parts. The branches of mathematics, mainly set theory, measure theory and analysis, define different terms regarding this application of partition. The author adds another term, the "Coarsity of Compression Model Hypothesis" (CCMH). This hypothesis is the assumption that all (a priori) proportionately partitioned parts can always be partitioned proportionately again, as a generalization. In order to explain the CCMH, the author will regard types of continuum and the general view of scales as perspectives. Finally, the author will define the digital measure constant as a continuum relation.

While the Cartesian Coordinate System is the human mind's intuition to partition physical space into patterns in order to recognize physical space itself, the term "intuitively equal" is used rather than "equal" because the real number set (the real number line) is uncountable in terms of counting numbers. Via non-standard analysis one can construct the same a priori idea of partition on the unit interval, and then apply this idea of partition to unidentified continuous functions. Work of this type has been done before in Physics Letters A by R. F. Werner and M. P. H. Wolff [21].

Revised version, 31 October 2026. This version keeps the original argument and states its results in precise form: $Hyp_1 = 1-\gamma$ as the relative displacement between x and its digital version, its finite-step form, the unit relation behind Hyp_0, and the photon energy deficit.

A. *Types of Continuum for Unidentified Continuity*

We begin with the idea of certain graphs (images) which look identical when looked at from far away. Since a "function" is the representation of an image on a graph, the distinctions between small-scale models of *"functions that look identical from far away"* will be separated into two kinds of continuum. This analysis was made in the article *"Tension between Intuitive Infinitesimals and Formal Mathematical Analysis"* [19] by Mikhail G. Katz and David Tall. The author mainly changed Luzin's saw (sawtooth) into ceil/floor series as a generalization.

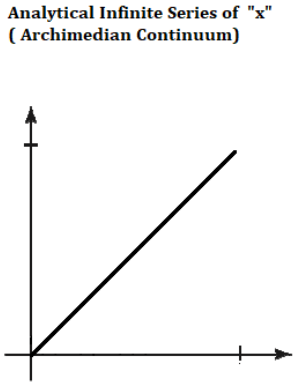


Fig. 2. Analog Continuum Model as x

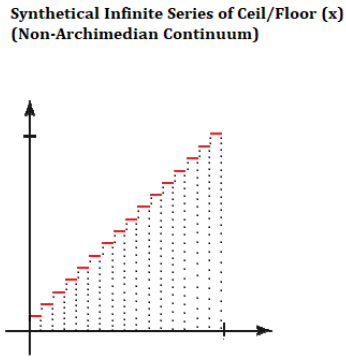


Fig. 3. Digital Continuum Model as $\lceil x \rceil$

In Figures 2–3, the two graphs (images) of functions look identical when looked at from a far distance. For the generalization, the term "unidentified" will be used for functions f (see below) that look like generalizations of an infinitely differentiable function (n indicates infinite) when looked at from far away (large scale), but are "unknown" in terms of continuum (small scale):

$$\frac{\partial^n f}{\partial x^n} \cong 0\,; \qquad f(x) \cong \sum_{i\in\mathbb{N}} b_i \cdot x^i\,; \quad b_i \in \mathbb{R}$$

Here and below, $A \cong B$ means that $A - B$ is infinitesimal, that is, A and B are equal when looked at from large scale.

Any area below the curve of the two possible continuum models of an "unidentified" function will also look like the same constant number when looked at from far away. But in reality there will be a difference between the defined continuum models. The difference between a value x and its digital version is the fractional part $\{x\}$: a sawtooth whose teeth keep their shape however small the steps become. This is Luzin's saw, generalized here to steps of any size.

The reason why ceil/floor series are used for the digital (non-Archimedean) continuum model, rather than the functional representations described in *"Tension between Intuitive Infinitesimals and Formal Mathematical Analysis"* at pages 14–15, is that the *"Natural Number Set"* representation as *"intuitively equal"* interval partitions can be shown with ceil/floor series.

II. Generalized Functional Model Interpretations

The abstraction is for the functional integration of natural numbers (as groups) with real numbers. Any continuum model of an "unidentified" function f at small scale is therefore shown with these functional representations. For a step $dx > 0$, write the floor and ceil at that step, and the displacement between x and its digital version, as

$$\lfloor x \rfloor_{dx} = dx \left\lfloor \frac{x}{dx} \right\rfloor, \qquad \lceil x \rceil_{dx} = dx \left\lceil \frac{x}{dx} \right\rceil, \qquad \{x\}_{dx} = x - \lfloor x \rfloor_{dx}$$

The displacement $\{x\}_{dx}$ is a sawtooth: it rises with the analytical model on every step and drops back at every step boundary. The author will regard the relative displacement $\{x\}_{dx}/x$, the gap between digital and analog as a fraction of the size, and measure it with dx/x, the measure on the positive numbers that does not change under scaling. The constant Hyp_1 is the representation of this total relative displacement, taken from the first step to a potential infinity H:

$$\int_{dx}^{H} \frac{\{x\}_{dx}}{x} \cdot \frac{dx}{x} \;\cong\; Hyp_1$$

The same constant appears in the ceil and floor models of the function $1/x$, whose two gaps fill exactly one unit:

$$\int_{dx}^{H} \Big(\frac{1}{x} - \frac{1}{\lceil x \rceil_{dx}}\Big) dx \cong Hyp_1\,, \qquad \int_{dx}^{H} \Big(\frac{1}{\lfloor x \rfloor_{dx}} - \frac{1}{x}\Big) dx \cong 1 - Hyp_1$$

All three integrals have the same value at every step dx (Theorem 1).

Note: The integration is taken on the domain $[0, H]$ because the generalized functional model is applied from a small-scale perspective. Any definite integration of length, area or volume that results in a constant number from the large-scale perspective is assumed to be "potentially infinite" in small-scale models. Here H is such a potential infinity: an infinite hyperreal bound, so that the integral is a well-defined element of $\mathbb{R}^*$. Hence the representation of the digital continuum is possible regardless of the basis

domain. The term "actual infinite" (∞) will be used as the "singular infinite", and "potential infinite" (H) as a perspective. At the scale dx, ordinary numbers such as 1, 2, 3 are themselves infinite: the number 1 is Hyp_0 steps and 2 is $2Hyp_0$ steps (Section IV). A digital measure constant must therefore have the same value, in the sense of $\cong$, for every infinite bound H.

III. Model Interpretations on Small Scale

The abstraction is to complement the smaller-scale perspective. The symbol $(dx)_{\|\mu\|}$ will indicate the digital (non-Archimedean) continuum at small-scale perspective. The digital continuum $(dx)_{\|\mu\|}$ has an interrelation with ceil/floor series as a functional representation. This is mainly indicated via the hyperreal set, shown with the symbol $\mathbb{R}^*$. (Here the author must cite Hilbert [8]–[10], Hausdorff [6], [7], Lebesgue [11], Borel [12], [13] and Robinson [15], because of various foundations and relations in analysis, set theory and measures.)

$$0 < a < +\infty\,; \qquad a = n \cdot (dx)_{\|\mu\|}$$

$$a \in \mathbb{R}^*\,; \quad n \in \mathbb{N}^+\,; \qquad \frac{\pm 1}{(dx)_{\|\mu\|}} \triangleq \pm\infty$$

Here the non-integer coefficient variants of $(dx)_{\|\mu\|}$ unit/group lengths are non-definitive for the $(dx)_{\|\mu\|}$ units/groups. So the CCMH is basically an "Intended Interpretation" model on the unit interval.

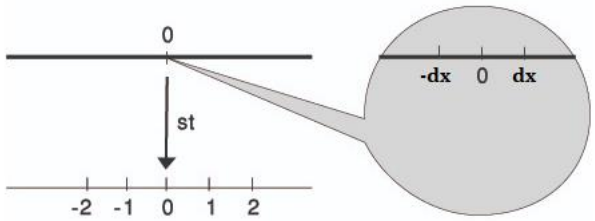


Fig. 4. A proportionate partitioning

IV. Digital Measure Constant

On earlier pages the statement *"Any area below the curve of the two possible continuum models of an unidentified function will also look like the same constant number when looked at from far away. But in reality there will be a difference between the defined continuum models"* was given. By area, this difference vanishes at large scale; by relative displacement, it does not. The author will regard the type of "unidentified" function f with the term

$$\Delta x \cdot \int_{\Omega_1} f$$

for functions that are variations of analytical models, which have an Archimedean continuum at small scale. The author will also regard the type of "unidentified" function f from a closer view with the term

$$\int_{\Omega_1} f\,dx$$

for functions that are variations of synthetical models, which have a digital (non-Archimedean) continuum at small scale. Precisely, $\Delta x \cdot \int_{\Omega_1} f$ is the integral of the analytical model, and $\int_{\Omega_1} f\,dx$ is the step sum of the synthetical model, $\sum_k f(\xi_k)\,dx$, taken over cells of length dx with $f(\xi_k)$ the upper (or lower) value of f on each cell.

Since any finite area below the curve of an "unidentified" function in the large-scale view looks non-finite when looked at from the small-scale view, the area difference between the models can be shown as nearly equal to zero in the large-scale view.

If on $\Omega_1 := [a,b]$ the large-scale integration of f is known to be a finite constant S,

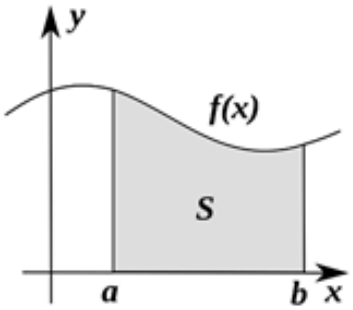


Fig. 5. Area under the curve of unidentified function

then the equations below must hold for the large-scale view:

$$\Delta x \cdot \int_{\Omega_1} f \cong S\,, \qquad \int_{\Omega_1} f\,dx \cong S\,, \qquad S \in \mathbb{R}$$

$$\left|\Delta x \cdot \int_{\Omega_1} f - \int_{\Omega_1} f\,dx\right| \cong 0$$

Note: If the domain or range of the function is extended to infinity anyhow, it must be extended to infinity regardless of all perspectives. Here the functional representation of "proportion" is entirely due to the perspective; therefore a potential point resides *at some infinity, not "the infinity" itself* [14]. This is due to Carathéodory's extension theorem (lemma):

$$\mu^*(S) = \mu^*(S \cap M) + \mu^*(S \cap M^c)$$

However, the two continuum models are not distinguished by area alone. They are distinguished by the displacement between a value and its digital version, which keeps its sawtooth shape at every scale. Measured relative to the size, with the scale-invariant measure, this displacement does not vanish at large scale and does not depend on which infinity bounds the integration. Hence the difference between the models results in the

digital measure constant Hyp_1. Hyp_1 is a constant because it determines the measurability of the two models. It cannot be equal to zero, since that would mean the two continuum models are equal. And it must be a constant, since if the difference between the two possible continuum models did not hold as a constant, definite (finite) measurements in the large-scale view would give different results for the two models because of the expansion of the compared maps (graphs, images), which would contradict large-scale definite integration (Gromov [16]). The value of this constant is given precisely as follows.

Theorem 1: Let $dx > 0$ and $N \in \mathbb{N}^+$, $N \geq 2$. Then, for every step dx,

$$Hyp_1(N) := \int_{dx}^{N\,dx} \frac{\{x\}_{dx}}{x^2}\,dx = 1 + \ln N - H_N\,, \qquad H_N = \sum_{k=1}^{N} \frac{1}{k}$$

Consequently, for every infinite hypernatural N,

$$Hyp_1 \;\cong\; 1-\gamma \;\approx\; 0.4227843351$$

where γ is the Euler–Mascheroni constant, $\gamma \cong H_N - \ln N$.

Proof: On the cell $[k\,dx, (k+1)\,dx]$, $\{x\}_{dx} = x - k\,dx$, and

$$\int_{k\,dx}^{(k+1)\,dx} \frac{x - k\,dx}{x^2}\,dx = \ln\frac{k+1}{k} - \frac{1}{k+1}$$

with no dx left. Summing over $k = 1, \ldots, N-1$ telescopes to $\ln N - (H_N - 1)$. For infinite N, $H_N - \ln N \cong \gamma$. ■

Theorem 1 gives the claimed properties of Hyp_1 exactly:

1) *Independent of the step:* $Hyp_1(N)$ contains no dx. The ceil and floor gaps of $1/x$ in Section II are likewise $1-\gamma$ and γ at every step, and together fill exactly one unit.
2) *The same for every infinity:* for infinite N and M, $Hyp_1(N) - Hyp_1(M) \cong 0$. Integrating up to 1, 2, 3 or any ordinary number, each of which is infinite at the scale dx, gives the same constant.
3) *Non-zero:* $Hyp_1 = 1 - \gamma \neq 0$, so the two continuum models are not equal. The displacement vanishes only where the digital and analog values coincide.
4) *Unique:* for a power weight x^{-p}, substituting $x = dx \cdot t$ gives $\int_{dx}^{N\,dx} \{x\}_{dx} x^{-p}\,dx = dx^{2-p} \int_1^N \{t\}\, t^{-p}\,dt$. The value is independent of the step only for $p = 2$, so the relative displacement with the scale-invariant measure is the only such weighting that yields a constant.
5) *Finite steps:* since $H_N = \ln N + \gamma + \frac{1}{2N} - \frac{1}{12N^2} + \ldots$,

$$Hyp_1(N) = (1-\gamma) - \frac{1}{2N} + \frac{1}{12N^2} - \ldots$$

Remark (other measures): Other measures of the gap between the two models need a normalization, and so depend on the chosen infinity. The area difference per step is $\frac{1}{2}\,dx \cdot V_a^b(f)$, and the ratio of staircase length to curve length is bounded by $\sqrt{2}$; their totals grow with H. Only the relative displacement of Theorem 1 converges without normalization. More generally,

$$\int_1^\infty \{x\}\, x^{-s-1}\,dx = \frac{1}{s-1} - \frac{\zeta(s)}{s}$$

so the digital–analog constants are values of the Riemann zeta function, and $1-\gamma$ is the scale-invariant case, reached as $s \to 1$. Since $\{x\} - \frac{1}{2} = -\sum_{n\geq 1} \sin(2\pi n x)/(\pi n)$, the displacement is a superposition of frequencies, which connects the construction to time–frequency analysis.

The two perspectives can also be compared through the integral itself. There is a special case when f is dx. This is because dx acts as a *constant* on $\mathbb{R}$, because $\mathbb{R}$ is *Archimedean*, and acts as an *infinitesimal* on $\mathbb{R}^*$, because it is *non-Archimedean*:

$$f(x) \cong dx$$

The integral therefore has to abide by two situations. The first is when f acts as the *constant number* $dx > 0$, with the actual infinity as upper bound:

$$\int_0^\infty dx \cdot dx = dx \cdot \int_0^\infty dx = \infty$$

The equations above hold since the large-scale integration of this integral is infinite. The second is when f acts as the *infinitesimal number* dx. The upper bound is then the potential infinity of the small-scale perspective, $H = 1/dx$:

$$\frac{\pm 1}{(dx)_{\|\mu\|}} \triangleq \pm\infty$$

$$\int_0^H dx = \int_0^{1/dx} dx = \frac{1}{dx} = 1 + 1 + 1 + \ldots$$

where the sum has $1/dx$ terms. Therefore

$$\int_0^H dx \cdot dx = dx \cdot \int_0^{1/dx} dx = 1$$

The two situations use different upper bounds, the actual infinity ∞ and the potential infinity $H = 1/dx$, which is why they give different values. This special case does not measure the displacement of Theorem 1: a constant function has the same value in both models, so it shows no displacement. Instead it compares the two perspectives. In the analytical perspective one unit of length is Δx; in the small-scale perspective the same unit is $dx \cdot (1/dx)$ steps. Their ratio defines the second constant:

$$\frac{\Delta x}{dx} = Hyp_0$$

that is,

$$\Delta x = Hyp_0 \cdot dx$$

Meaning: any unit distance Δx on $\mathbb{R}$ contains Hyp_0 steps of the digital continuum $\mathbb{R}^*$. Hyp_0 is the conversion factor between the two units, like the number of centimetres in a metre. With $\Delta x = 1$, $Hyp_0 = 1/dx$ is an infinite hypernatural number. In order to obtain the values of the digital measure constants Hyp_0 and Hyp_1, the author indicates that the generalized functional model interpretations of Section II lead to the result: the relative displacement between x and its digital version is the Hyp_1

constant, $Hyp_1 \cong 1-\gamma$ (Theorem 1). The two constants are connected when the displacement is measured over one unit length Δx, which contains Hyp_0 steps:

$$Hyp_1(Hyp_0) = 1 + \ln Hyp_0 - H_{Hyp_0} \;\approx\; (1-\gamma) - \frac{1}{2\,Hyp_0}$$

If the continuum is analog, Hyp_0 is infinite and $Hyp_1 = 1-\gamma$ in the sense of $\cong$. If the continuum is digital with a finite number of steps, Hyp_1 falls short of $1-\gamma$ by $1/(2Hyp_0)$. Conversely, the shortfall determines the number of steps:

$$Hyp_0 \;\approx\; \frac{1}{2\left[(1-\gamma) - Hyp_1(Hyp_0)\right]}$$

V. Digital Measure

Any function that represents a generalized signal and has unidentified continuity shown on the real numbers $\mathbb{R}$ (far-away view) can also be shown on the hyperreal number set $\mathbb{R}^*$. The author indicates the existence of another digital measure constant, Hyp_0, via generalization, in order to show that a point at real coordinate x lies at digital coordinate (step count) x^* with

$$x^* = Hyp_0 \cdot x\,; \qquad x \in \mathbb{R}\,; \quad x^* \in \mathbb{R}^*$$

The digital measure of an interval $[a,b]$ on $\mathbb{R}$ is its number of steps:

$$\mu_D[a,b] = Hyp_0 \cdot (b-a)\,; \qquad b > a$$

Equivalently, an interval $[a^*,b^*]$ given in digital coordinates has real length

$$\frac{1}{Hyp_0} \cdot (b^* - a^*)$$

VI. Applications

If there is a signal f (shown via a function) that is known to have unidentified continuity,

$$\frac{\partial^n f}{\partial x^n} \cong 0\,; \qquad f(x) \cong \sum_{i\in\mathbb{N}} b_i \cdot x^i\,; \quad b_i \in \mathbb{R}$$

then via this signal one can obtain the digital continuum model or the analog continuum model of this signal. So any unidentified continuous function f can be transferred from one continuum model to the other by

$$f(x) = f\left(\frac{x^*}{Hyp_0}\right)\,; \qquad x \in \mathbb{R}\,; \quad x^* \in \mathbb{R}^*$$

An example is the energy of the photon (wave–particle), shown via functions ($E(x)$, $f(x)$ and $\lambda(x)$ for the functional representations of energy, frequency and wavelength) that have unidentified continuity:

$$E(x) = h \cdot f(x)\,; \qquad f(x) = \frac{c}{\lambda(x)}\,; \qquad \lambda(x) = x$$

Then the digital continuum model of wavelength (as an interval) is

$$\lambda = \frac{\lambda^*}{Hyp_0}\,; \qquad \lambda^* \in \mathbb{R}^*\,; \quad \lambda \in \mathbb{R}$$

with the bounded range

$$\lambda^* \in \mathbb{R}^*\,; \qquad 0 \le \lambda^* \le \frac{1}{dx}$$

If every wavelength is an integer multiple of the interval $[0,dx]$, then $\lambda^* = n$ with $n \in \mathbb{N}^+$ and $n \le N$, where $N = 1/dx$. The energies are then

$$E_n = h \cdot \frac{c}{\lambda} = \frac{h\,c\,Hyp_0}{n}\,; \qquad n = 1,2,\ldots,N$$

These energy levels are not integer multiples of each other; they form the sequence $E_1, E_1/2, E_1/3, \ldots$ Nevertheless, a single *logical* unit u exists such that every level is an integer multiple of it. Let $L = \mathrm{lcm}(1,2,\ldots,N)$, which exists as a hypernatural number by transfer. Then

$$u = \frac{h\,c\,Hyp_0}{L}\,; \qquad E_n = \frac{L}{n} \cdot u\,, \quad \frac{L}{n} \in \mathbb{N}^+$$

For example, for a finite range with $N = 10$, $L = 2520$ and the levels are $2520u, 1260u, 840u, \ldots$ The bounded range of λ^* is essential: without an upper bound on n, no single unit divides every level. The author therefore indicates that if every wavelength in a bounded range is an integer multiple of the interval $[0,dx]$, then every energy level is an integer multiple of the logical constant u. This holds within the digital continuum model and is not a statement about measured photon spectra.

The same levels connect to both constants. First, take $\Delta x = 1$, so that $N = Hyp_0$. Since $E_1/u = \mathrm{lcm}(1,\ldots,Hyp_0)$, and the logarithm of $\mathrm{lcm}(1,\ldots,N)$ is the Chebyshev function $\psi(N)$, with $\psi(N)/N \to 1$ (equivalent to the prime number theorem),

$$\psi(Hyp_0) = \ln \frac{E_1}{u}\,, \qquad Hyp_0 \approx \ln \frac{E_1}{u}$$

Second, the levels $E_n = E_1/n$ are the function $1/x$ of Section II. Comparing the digital spectrum, a sum over the levels, with the analog spectrum, an integral over n:

$$\int_1^{Hyp_0} \frac{E_1}{n}\,dn - \sum_{n=2}^{Hyp_0} E_n = E_1\left(1 + \ln Hyp_0 - H_{Hyp_0}\right) = E_1 \cdot Hyp_1(Hyp_0)$$

$$\sum_{n=1}^{Hyp_0} E_n - \int_1^{Hyp_0} \frac{E_1}{n}\,dn \;\cong\; \gamma E_1$$

In a continuum, Hyp_0 is infinite and the relative deficit is exactly $Hyp_1 = 1-\gamma$. In a discrete world with Hyp_0 steps per unit length, Theorem 1 gives the deficit to leading order. Since $E_1/(2Hyp_0) = hc/(2\Delta x) = \frac{1}{2}E_{\min}$, where $E_{\min} = hc/\Delta x$ is the lowest level (wavelength equal to the unit length),

$$\text{deficit} \;\approx\; (1-\gamma)E_1 - \tfrac{1}{2}E_{\min}$$

The correction does not depend on dx: it is half of the lowest energy quantum.

Wavelengths	Relative deficit	Absolute deficit
Continuous	$1-\gamma$	$(1-\gamma)E_1$
Discrete (Hyp_0 steps)	$(1-\gamma)-\frac{1}{2Hyp_0}$	$(1-\gamma)E_1-\frac{1}{2}E_{\min}$

$E=hf$ holds in both cases; what differs is whether the frequencies take every value or only the values $c/(n\,dx)$. Which row the spectrum follows is therefore a test between the analog and the digital continuum model. In the continuous case only the relative form is meaningful, since $E_1=hc/dx$ has no finite value without a smallest step.

VII. Conclusion

Within the non-Archimedean continuum model, the digital measure constant exists and has a single value. The displacement $\{x\}$ between x and its digital version, measured relative to x with the scale-invariant measure dx/x, gives $Hyp_1 \cong 1-\gamma \approx 0.4228$. This value does not depend on the step and is the same for every infinite bound, as required because ordinary numbers are infinite at the scale dx. Over one unit length, which contains Hyp_0 steps, $Hyp_1=(1-\gamma)-1/(2Hyp_0)$ to leading order: the shortfall $1/(2Hyp_0)$ is zero for an analog continuum and positive for a digital one. In the photon spectrum this becomes a deficit of $(1-\gamma)E_1$ in a continuum and $(1-\gamma)E_1-\frac{1}{2}E_{\min}$ in a discrete world. The constant $Hyp_0=\Delta x/dx$ converts between the real and the digital units of length and is tied to the energy levels by $\psi(Hyp_0)=\ln(E_1/u)$.

Data Availability Statement

Some of the data is available to the public, while some of the data has prerequisites, and some of the data is located at an outdated website. The author has given information about the contents he has read below. All referenced pages are at the equivalent of the internet domain's reference pages (logical), not the original book's or reference's pages (physical).

References and Footnotes

[1] Riemann, Bernhard, *Gesammelte mathematische Werke und wissenschaftlicher Nachlass*, pages: (18–58), (78–151), (152–159), (228–265), (269–283), (298–330), (346–359), (372–419), (448), (464–470). https://archive.org/details/bernardrgesamm00riemrich/page/n17/mode/2up?view=theater

[2] Lie, Sophus, *Theorie der Transformationsgruppen*, Reihe 1, Seite: (III–VIII), (1–8), (11–136), (171–183), (184–211), (289–310), (554–597). https://gdz.sub.uni-goettingen.de/id/PPN585383391

[3] Lie, Sophus, *Theorie der Transformationsgruppen*, Reihe 2, Seite: (35–44), (77–114), (171–176), (178–206), (292–293), (298–327). https://gdz.sub.uni-goettingen.de/id/PPN588388580

[4] Lie, Sophus, *Theorie der Transformationsgruppen*, Reihe 3, Seite: (IX–XXV), (1–109), (122–170), (180–232), (266–282), (296–398), 427, (437–471), (485–497), (524–635), (665–753). https://gdz.sub.uni-goettingen.de/id/PPN588388718

[5] Cantor, Georg, *Gesammelte Abhandlungen* (ed. Ernst Zermelo), Seite 115 (§2). https://gdz.sub.uni-goettingen.de/id/PPN237853094

[6] Hausdorff, Felix, *Grundzüge der Mengenlehre*: (73), (101–139), (158–161), (185–189), (209–219), (229–239), (247–249), (253–335), (369–398), (403–417), (430–448). https://gallica.bnf.fr/ark:/12148/bpt6k9736980b

[7] Hausdorff, Felix, *Gesammelte Abhandlungen, Band 3: Deskriptive Mengenlehre und Topologie*, U. Felgner, H. Herrlich, M. Hušek, V. Kanovei, P. Koepke, G. Preuß, W. Purkert, E. Scholz (eds.), Springer: (27–62), (91–341), (374–430), (468–472), (483–486), 494–497, (543–548), 559, (725–745), (757–776), (785–802), (805–832), (847–899), (909–987).

[8] Hilbert, David, *Gesammelte Abhandlungen*, Band I, Seite: (5–23), (26–42), (53–68), (69–92), (112–149), (176–242), (275–279), (291–302), (390–393), (412–427), (507–509). https://gdz.sub.uni-goettingen.de/id/PPN237821559

[9] Hilbert, David, *Gesammelte Abhandlungen*, Band II, Seite: (1–24), (34–37), (102–147), (162–263), (287–319), (341–344), (376–414). https://gdz.sub.uni-goettingen.de/id/PPN237833719

[10] Hilbert, David, *Gesammelte Abhandlungen*, Band III, Seite: (1–25), (33–37), (43–80), (94–145), (290–377). https://gdz.sub.uni-goettingen.de/id/PPN237834022

[11] Lebesgue, Henri, *Leçons sur l'intégration et la recherche des fonctions primitives*, pages: (1–14), (17–35), (37–140), (156–184), (202–226), (262–271), (314–340). https://gallica.bnf.fr/ark:/12148/bpt6k377935x/f359.item

[12] Borel, Émile, *Leçons sur la théorie des fonctions*, pages: (20–81). https://archive.org/details/leonssurlathori00boregoog/page/n19/mode/2up

[13] Borel, Émile, *Leçons sur les séries divergentes*, pages: (83–113), (147–163). https://archive.org/details/leonssurlessrie01boregoog/page/n114/mode/2up

[14] Carathéodory, Constantin, *Mass und Integral und ihre Algebraisierung*, Birkhäuser, Seite: (10–172), (192–200), (247–281), (285–287), (294–310).

[15] Robinson, Abraham, *Non-Standard Analysis* (first edition), pages: (24–64), (65–118), (132–149), (189–192), (193–245).

[16] Gromov, Michael, "Groups of polynomial growth and expanding maps (with an appendix by Jacques Tits)," *Publications Mathématiques de l'IHÉS*, Tome 53 (1981), pp. 53–78. http://www.numdam.org/item/PMIHES_1981__53__53_0

[17] Ehrlich, Philip, "The rise of non-Archimedean mathematics and the roots of a misconception I: the emergence of non-Archimedean systems of magnitudes," pages: (16), (26–55). https://link.springer.com/article/10.1007/s00407-005-0102-4

[18] Katz, Mikhail G., Hrbacek, Karel, Kanovei, Vladimir, Fletcher, Peter, Lobry, Claude, Sanders, Sam, "Approaches to analysis with infinitesimals following Robinson, Nelson and others," page: (4). https://arxiv.org/abs/1703.00425

[19] Katz, Mikhail G., Tall, David, "Tension between intuitive infinitesimals and formal mathematical analysis," pages: (14–15). https://arxiv.org/abs/1110.5747

[20] Soto-Bajo, Moisés, Fraguela Collar, Andrés, Herrera-Vega, Javier, "On the concept of frequency in signal processing: a discussion

[Perspectives].” https://ieeexplore.ieee.org/document/10243508

[21] Werner, R. F., Wolff, M. P. H., “Classical mechanics as quantum mechanics with infinitesimal $\hbar$,” *Physics Letters A*. https://www.sciencedirect.com/science/article/abs/pii/0375960195003443